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Secondary hyperparathyroidism (SHPT) is a condition characterized by excessive PTH release in response to depressed serum calcium or elevated serum phosphate levels. The novel peptide AMG 416, an hCaSR agonist, is being investigated as a possible treatment for SHPT caused by end-stage renal disease (ESRD). IP-1, a metabolite that acts downstream of IP₃, is a marker of hCaSR activity. Researchers used embryonic kidney cells lacking hCaSR (HEK-293T) and HEK-293T clone cells stably transfected with hCaSR (hCaSR) to measure IP-1 levels following AMG 416 treatment (Figure 1).

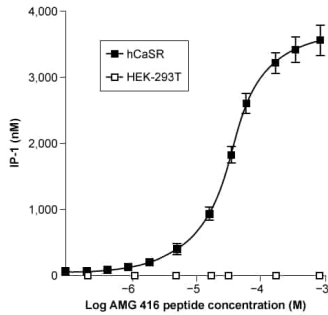


Figure 1 IP-1 concentration in embryonic kidney cells lacking hCaSR (HEK-293T) and HEK-293T clone cells transfected with hCaSR (hCaSR) following AMG 416 treatment

The dependence of AMG 416 on extracellular calcium was investigated by incubating hCaSR cells with increasing concentrations of AMG 416 in the presence or absence of calcium (Figure 2).

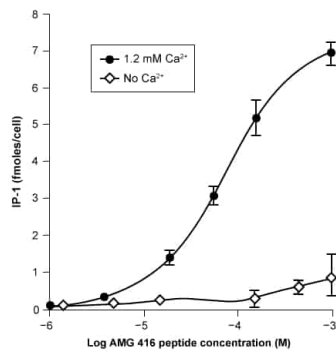


Figure 2 IP-1 levels in hCaSR cells incubated with and without calcium and AMG 416 peptide (Note: IP-1 levels are normalized to the number of cells present in each culture.)

In addition, the effect of AMG 416 on PTH concentration was measured in wild-type (WT) rats. Blood samples were taken from male and female rats for 8 hours following administration of varying concentrations of AMG 416 (Figure 3), and PTH levels were measured.

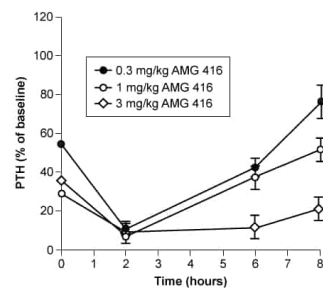


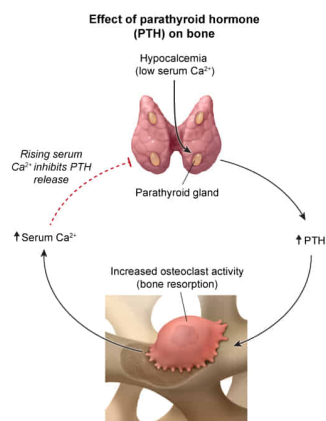
Figure 3 Serum PTH concentration following administration of AMG 416 to WT rats

Which of the following is a potential complication of secondary hyperparathyroidism?

- ✓ ☒ A. Decreased bone mineralization [67%]
☐ B. Decreased osteoclast activity [7%]
☐ C. Increased osteoblast activity [18%]
☐ D. Increased cortical bone mass [7%]

Correct
 67% answered correctly

Explanation:



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The **parathyroid glands** are four small endocrine glands located in the neck on the dorsal surface of the thyroid gland. These pea-sized glands produce parathyroid hormone (PTH), which controls plasma calcium homeostasis via a simple endocrine reflex. **PTH is released** in response to **decreased plasma calcium** and acts on bone, the kidneys, and the small intestine to produce a net increase in calcium. This increase in plasma calcium concentration serves as **negative feedback** and turns off the reflex, ending PTH secretion.

Based on the passage, secondary hyperparathyroidism (SHPT) is characterized by increased PTH secretion, which increases plasma calcium concentration. **PTH release** indirectly increases **bone resorption (breakdown)** by **osteoclasts**, bone-resorbing cells, and causes **decreased bone mineralization**. The majority of calcium in the body is stored as hydroxyapatite $[\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2]$, a mineral found in the bone matrix that primarily contributes to bone strength and hardness. Osteoclast-mediated bone destruction (ie, resorption) causes the calcium stored in the matrix to be released into the blood, dissolving mineralized bone and increasing plasma calcium levels.

(Choice B) Osteoclast activity is indirectly increased, *not decreased*, by PTH.

(Choice C) Osteoblasts are bone-forming cells that synthesize the organic and mineralized components of bone. Because the main function of PTH is to increase serum calcium levels (eg, via increased release of calcium from bone), the excessive PTH release characteristic of SHPT would likely increase osteoclast (*not* osteoblast) activity.

(Choice D) The dense outer layer of bone is referred to as the cortex, and the matrix of cortical (**compact/hard**) bone contains hydroxyapatite. Prolonged indirect osteoclast activation due to PTH secretion dissolves hydroxyapatite, releasing the mineral components into the blood and *decreasing* cortical bone mass.

Educational objective:

The main effect of parathyroid hormone (PTH) activity is to increase serum calcium concentration. PTH secretion from the parathyroid gland is triggered by decreased plasma calcium. PTH indirectly stimulates osteoclast activation, promoting bone breakdown and the release of ionic calcium into the blood.

Time spent: 0 sec

Subject:
Biology

QID: 400974

System:
Endocrine and Nervous Systems

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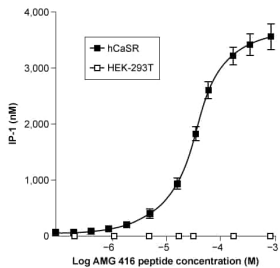


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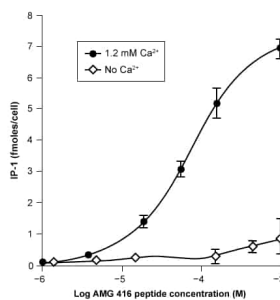


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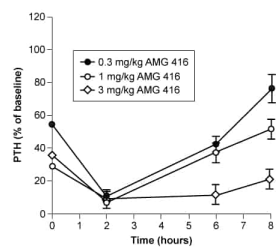
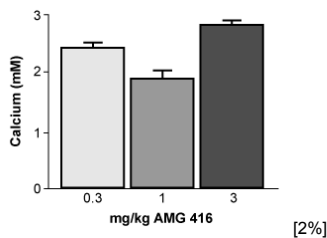


Figure 3 Serum PTH concentration following administration of AMG 416 to WT rats

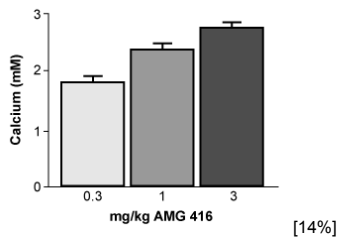
In another experiment, calcium levels were measured in WT rats exposed to different dosages of AMG 416. Assuming that AMG 416 treatment suppresses PTH release, which graphic represents the blood calcium levels that would be expected in these rats 8 hours after injection with indicated dosages of AMG 416?

- ✓ ☒ A.
-
- | mg/kg AMG 416 | Calcium (mM) |
|---------------|--------------|
| 0.3 | ~2.4 |
| 1 | ~2.1 |
| 3 | ~1.8 |
- [81%]
- ☐ B.
-
- | mg/kg AMG 416 | Calcium (mM) |
|---------------|--------------|
| 0.3 | ~2.4 |
| 1 | ~2.3 |
| 3 | ~1.8 |
- [1%]
- ☐ C.



[2%]

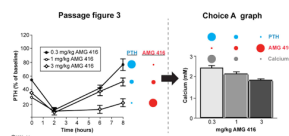
○ D.



[14%]

Correct
80% answered correctly

Explanation:



According to the passage, AMG 416 is an hCaSR agonist, meaning that AMG 416 is able to bind the hCaSR receptor in the parathyroid glands and induce a similar biological response as that of normal ligand binding (ie, inhibition of PTH release). Consequently, the question states that AMG 416 treatment suppresses PTH release.

Figure 3 shows that **as AMG 416 dosage increases, PTH levels decrease**. This graph depicts PTH concentration immediately and at three time points following administration of three different AMG 416 dosages to WT rats. Figure 3 shows that 8 hours after 0.3 mg/kg AMG 416 administration, PTH levels are highest. Treatment with 1 mg/kg AMG 416 results in moderate levels of PTH, and treatment with 3 mg/kg results in the lowest PTH levels after 8 hours. Therefore, PTH is *most suppressed* at the highest AMG 416 dosage (ie, 3 mg/kg) 8 hours following administration.

The net result of PTH suppression is **decreased circulating calcium** (ie, the opposite of PTH release). **Less PTH due to progressively higher AMG dosages causes greater decreases in serum calcium**. Therefore, as shown in the graph in Choice A, greater AMG 416 dosages correspond to decreased serum calcium concentrations (**Choices B, C, and D**).

Educational objective:

PTH release causes increased plasma calcium. Decreased PTH availability or diminished physiological response to PTH decreases calcium levels in the blood.

Time spent: 0 sec
Subject:
Biology

QID: 400975
System:
Endocrine and Nervous Systems

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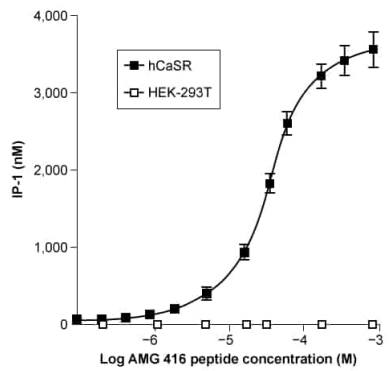


Figure 1 IP-1 concentration in embryonic kidney cells lacking hCaSR (HEK-293T) and HEK-293T clone cells transfected with hCaSR (hCaSR) following AMG 416 treatment

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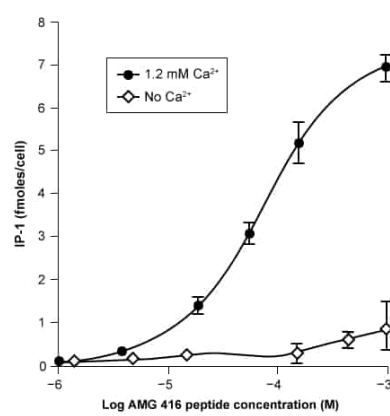


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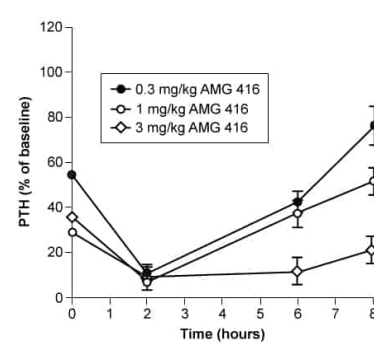


Figure 3 Serum PTH concentration following administration of AMG 416 to WT rats

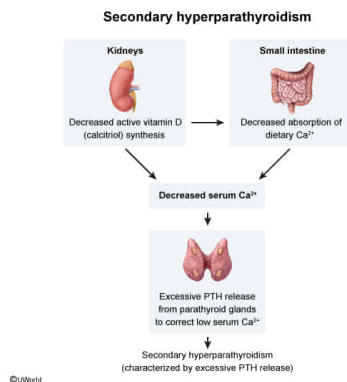
Secondary hyperparathyroidism may result from:

- ☐ A. decreased synthesis of active vitamin D and increased intestinal calcium absorption. [19%]
- ☐ B. increased synthesis of active vitamin D and increased intestinal calcium absorption. [18%]
- ☒ C. decreased synthesis of active vitamin D and decreased intestinal calcium absorption. [42%]
- ☐ D. increased synthesis of active vitamin D and decreased intestinal calcium absorption. [20%]

Correct

41% answered correctly

Explanation:



PTH increases the activity of the enzyme that catalyzes the final step in the conversion of inactive circulating vitamin D into its active form, calcitriol. Vitamin D can be absorbed from dietary sources or produced from its cholesterol precursor in a process requiring skin exposure to ultraviolet light. However, successive reactions must occur for vitamin D to become physiologically active. The first reaction occurs in the liver and the second, which is stimulated by PTH, occurs in the kidneys to create active 1,25-dihydroxyvitamin D (calcitriol).

The passage explains that secondary hyperparathyroidism (SHPT) involves increased PTH release in response to low serum calcium. **SHPT could result from deficient synthesis of active vitamin D (calcitriol)**, which in turn leads to **decreased intestinal calcium absorption**. These changes lead to low levels of circulating calcium and require constant PTH release to correct this imbalance, which may result in the development of SHPT.

(Choices A and B) Increased intestinal calcium absorption raises blood calcium levels, which serves as a negative feedback signal to end PTH secretion. As a result, increased intestinal calcium absorption would *not* cause the increased PTH release characteristic of SHPT.

(Choice D) Increased synthesis of active vitamin D results in increased (*not* decreased) intestinal calcium absorption, and therefore *inhibits* the release of PTH. This would *not* result in SHPT, a disease characterized by *excessive* release of PTH.

Educational objective:

PTH release stimulates the synthesis of the active form of vitamin D (calcitriol) in the kidneys. Consequently, active vitamin D primarily functions to promote absorption of dietary calcium from the small intestine.

Time spent: 0 sec

Subject:
Biology

QID: 400976

System:
Endocrine and Nervous Systems

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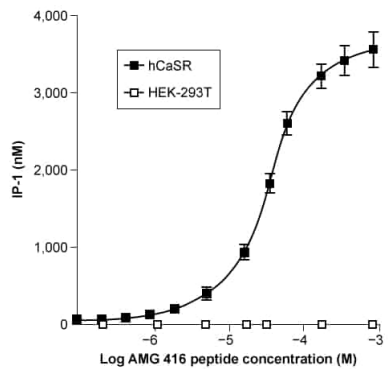


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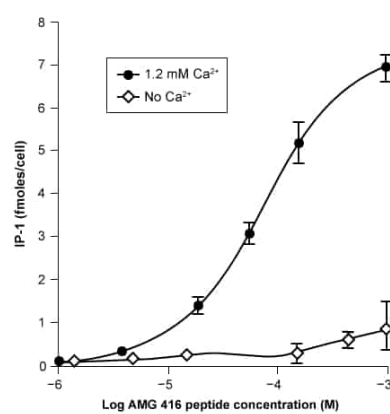


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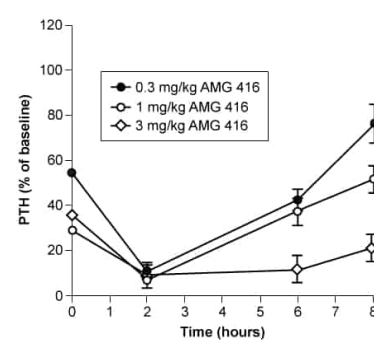


Figure 3 Serum PTH concentration following administration of AMG 416 to WT rats

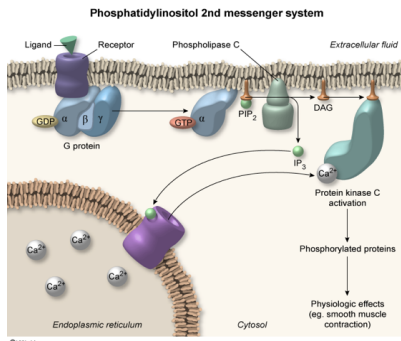
Based on the passage and the experimental results shown in Figure 1, which event is an outcome of treating hCaSR cells with AMG 416?

- ☐ A. PTH binding to the G protein alpha subunit [12%]
☒ B. GTP binding to the G protein alpha subunit [59%]
☐ C. Calcium binding to phospholipase C [17%]
☐ D. ATP binding to phospholipase C [10%]

Correct

58% answered correctly

Explanation:



The passage states that the hCaSR is a G protein–coupled receptor (GPCR) whose activity is marked by IP-1, a metabolite downstream of IP₃. This indicates that hCaSR specifically activates the IP₃–DAG pathway. In addition, AMG 416 is an *agonist* of hCaSR, meaning it functions to bind and *activate* hCaSR. As a result, the following processes occur following hCaSR activation by AMG 416:

1. The G protein heterotrimer (alpha, beta, gamma subunits) attaches to the intracellular portion of the GPCR, a transmembrane cell surface receptor. When the ligand binds to the extracellular domain of the receptor, the *GDP* molecule initially bound to the G protein alpha subunit is released, and a **GTP molecule binds the G protein** in its place.
2. The **GTP-bound alpha subunit** dissociates from the beta and gamma subunits and activates the membrane-bound enzyme phospholipase C (PLC).
3. PLC hydrolyzes the membrane phospholipid phosphatidylinositol biphosphate (PIP₂) into inositol trisphosphate (IP₃) and diacylglycerol (DAG).
4. Water-soluble IP₃ diffuses from the cell membrane through the cytoplasm to bind IP₃ receptors located on the endoplasmic reticulum and mitochondria. Binding of the IP₃ receptor opens calcium channels, allowing the release of stored calcium into the cytoplasm, which activates protein kinase C (PKC).
5. DAG diffuses within the membrane and activates PKC.
6. PKC phosphorylates downstream intracellular proteins to produce its physiologic effects.

(Choice A) The ligand of a GPCR binds to the *extracellular* region of the receptor. The G protein is coupled to the receptor on its *intracellular* side, and therefore cannot be bound by the ligand, which in this case is calcium or AMG 416 (*not* PTH).

(Choice C) Stored cellular calcium binds and activates PKC, *not* PLC.

(Choice D) PLC is activated by the GTP-bound alpha subunit of the G protein, *not* ATP.

Educational objective:

When the alpha subunit of a G protein is bound to GDP, the protein is inactive. Upon ligand binding, GTP binds in place of GDP, causing activation of phospholipase C (PLC). Consequently, PLC breaks down phosphatidylinositol biphosphate into diacylglycerol (DAG) and inositol trisphosphate (IP₃). IP₃ then triggers calcium release from the endoplasmic reticulum, and protein kinase C is subsequently activated by both DAG and calcium.

Time spent: 0 sec
Subject:
Biology

QID: 400977
System:
Endocrine and Nervous Systems

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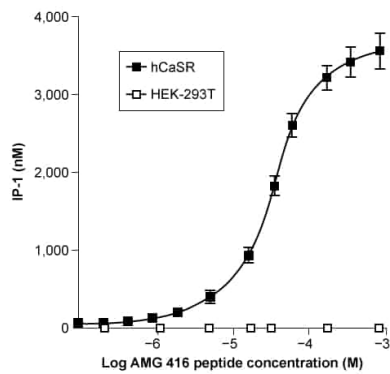


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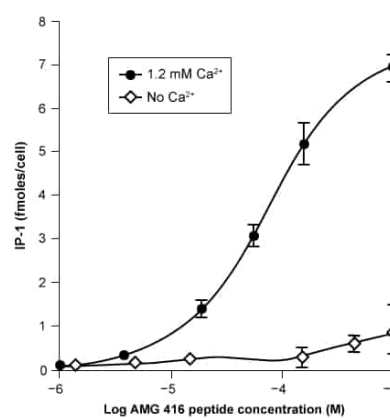


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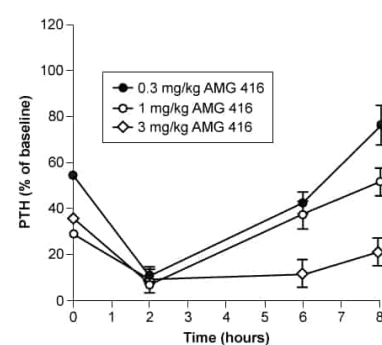


Figure 3 Serum PTH concentration following administration of AMG 416 to WT rats

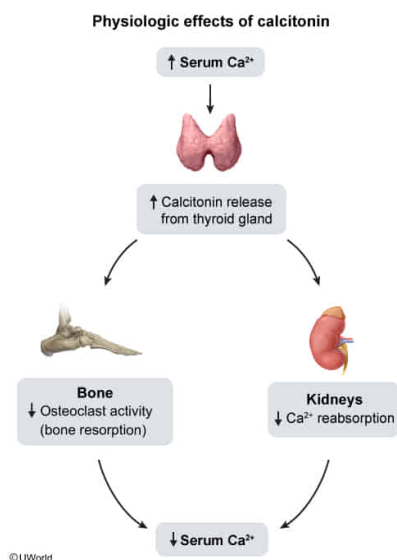
Researchers treated 5/6 Nx rats, animal models of SHPT in human ESRD, with regular intraperitoneal infusions of CaCl_2 to cause a sustained increase in serum calcium levels, which resulted in the release of a certain regulatory hormone. This regulatory hormone most likely originated in which structure?

- ☐ A. Thymus [6%]
- ☐ B. Pancreas [5%]
- ☐ C. Bone marrow [6%]
- ✓ ☒ D. Thyroid [81%]

Correct

81% answered correctly

Explanation:



In response to increased plasma calcium concentration, the peptide hormone **calcitonin** is secreted by the **thyroid gland**. Calcitonin mainly acts to reduce plasma calcium by **decreasing osteoclast activity (bone resorption)** and **increasing renal excretion of calcium**.

According to the question, 5/6 Nx rats are animal models of secondary hyperparathyroidism (SHPT) in human end-stage renal disease (ESRD). The passage states that a hallmark of SHPT is elevated PTH release due to insufficient circulating calcium. However, if increased serum calcium levels are sustained in these rats via regular CaCl_2 infusions, this physiological condition likely triggers **increased calcitonin release by the thyroid** in an attempt to lower plasma calcium and return to [homeostasis](#).

(Choice A) The thymus is a lymphoid structure that is the site of T lymphocyte maturation. T cell synthesis begins in the bone marrow but is not completed until lymphoid progenitor cells (T cell precursors) migrate to the thymus, where they mature. After this, the naïve T lymphocytes are released into circulation and are involved in [cell-mediated immunity](#), *not* calcium homeostasis.

(Choice B) The [pancreas](#) is a glandular organ that has both endocrine and exocrine functions. The exocrine pancreas secretes pancreatic juices containing digestive enzymes and bicarbonate, and the endocrine pancreas synthesizes the hormones insulin, glucagon, and somatostatin. However, neither pancreatic function regulates calcium homeostasis.

(Choice C) The [bone marrow](#) is the spongy, soft tissue found in bone cavities. [Hematopoiesis](#), or the synthesis of blood cellular components, occurs in the red marrow of flat bones and at the epiphysis of long bones. The yellow bone marrow is in the central cavity of long bones and is composed mostly of fat cells. Neither type of bone marrow is involved in calcium homeostasis.

Educational objective:

Calcitonin, a hormone synthesized by the thyroid gland, decreases calcium concentrations in the blood by inhibiting osteoclast activity (bone resorption) and promoting calcium excretion in the kidneys.

Time spent:0 sec
Subject:
Biology

QID:400978
System:
Endocrine and Nervous Systems

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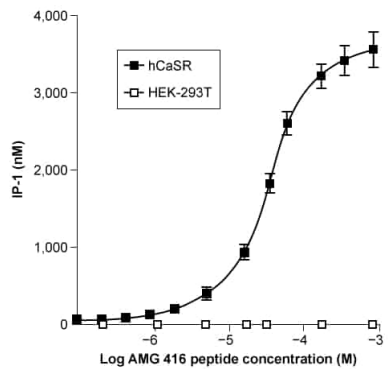


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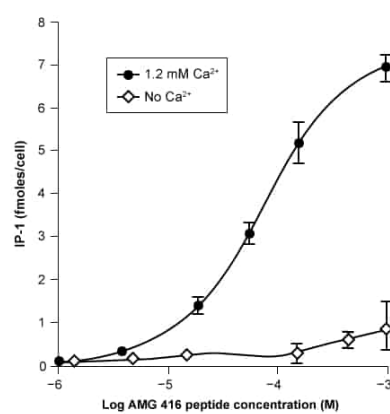


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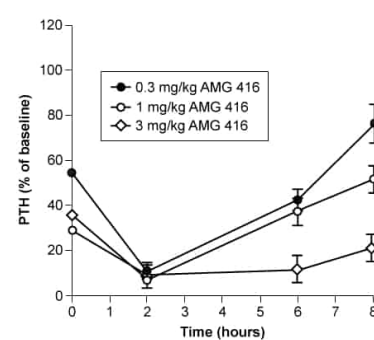


Figure 3 Serum PTH concentration following administration of AMG 416 to WT rats

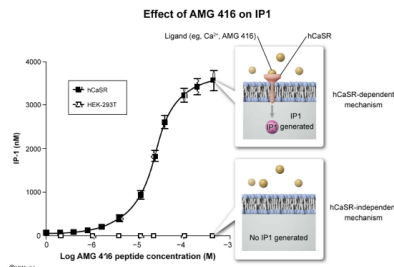
Which of the following conclusions is NOT consistent with the experimental findings presented in the passage?

- ☐ A. Calcium-free conditions resulted in the activation of hCaSR in hCaSR cells at higher AMG 416 dosages. [28%]
- ☐ B. Plasma PTH levels were suppressed to approximately the same level in WT rat cells 2 hours after administration of the different AMG 416 doses. [8%]
- ☒ C. AMG 416 increased cellular IP₃ levels through both hCaSR-dependent and hCaSR-independent mechanisms. [54%]
- ☐ D. Mean plasma PTH levels remained below baseline for at least 8 hours in the WT rat group treated with 3.0 mg/kg AMG 416. [8%]

Correct

54% answered correctly

Explanation:



The hCaSR is a G protein–coupled receptor and increases intracellular IP₃ when activated. The passage describes IP-1 as a metabolite of the second messenger IP₃ and as a marker of hCaSR activity. This means that IP-1 concentration is *directly correlated* with hCaSR activity.

The only option NOT consistent with the passage data is the conclusion that AMG 416 increased cellular IP₃ levels through both hCaSR-dependent and hCaSR-independent mechanisms. Figure 1 shows that AMG 416 increased IP-1 levels in cells expressing hCaSR. This increase concurs with the passage description of AMG 416 being a hCaSR agonist, a molecule that can bind a receptor and induce a biological response similar to that of ligand binding. However, Figure 1 also shows that treatment of cells lacking hCaSR (HEK-293T) resulted in *no change* in IP-1 concentration. Therefore, it is reasonable to conclude that **AMG 416 mediates increased intracellular IP₃ only through binding and activation of hCaSR** but NOT through an hCaSR-independent mechanism.

In contrast, the following conclusions *are* consistent with the data provided in the passage:

- Figure 2 shows that IP-1 levels begin to increase when hCaSR cells are incubated in the calcium-free condition (No Ca²⁺) and treated with AMG 416 peptide concentrations greater than 10⁻⁴ M. [Note: In the figure, this number is shown as -4, which is the result of calculating log₁₀(10⁻⁴).] Therefore, the passage data *supports* the conclusion that calcium-free conditions caused the activation of hCaSR in these cells at higher AMG 416 concentrations (**Choice A**).
- Figure 3 shows that plasma PTH levels were suppressed to approximately the same level in WT rat cells 2 hours after administering AMG 416 treatment at different dosages (ie, 0.3 mg/kg, 1.0 mg/kg, 3.0 mg/kg) (**Choice B**).
- Figure 3 also shows that PTH levels remained suppressed below baseline levels (ie, below 100%) for at least 8 hours in WT rats following AMG 416 treatment at 3.0 mg/kg (**Choice D**).

Educational objective:

An agonist enhances or duplicates the effect of a ligand binding a receptor. Agonists are able to bind a receptor and induce a biological response similar to that of normal ligand binding.

Time spent: 0 sec

Subject:
Biology

QID: 400979

System:
Endocrine and Nervous Systems